Lecture 6, Part I: Section A.6
Linear Inequalities

Properties of Inequalities

Inequalities involving absolute value

Common algebraic errors

F12 4th CAK

Lecture 6, Part I: Section A.6
Linear Inequalities

An **inequality** is a statement that two quantities are not equal.

To **solve** an inequality is to find all values of x that makes that the inequality true.

Unlike equations, most inequalities usually have an infinite number of solutions. To present the solution, we use **interval notation**.

Properties of Inequalities

Let a, b, c and d be real numbers.

1. Transitive Property If $a < b$ and $b < c$, then $a < c$
2. Addition of Inequalities

   If $a < b$ and $c < d$, then $a + c < b + d$
3. Addition/Subtraction of a Constant

   If $a < b$, then $a \pm c < b \pm c$
4. Multiplication/Division by a Constant

   For $c > 0$: If $a < b$, then $ac < bc$

   For $c < 0$: If $a < b$, then $ac > bc$ *

$2 < 7$

$-3(2) \quad 7(-3)$

$-6 > -21$

$$\overbrace{3(2)(6)}^{6\cdot 6} \neq 3(2+6)$$

$18 \cdot 2$

* **NOTE:** To multiply or divide both sides of an inequality by a **negative** number, **reverse** the inequality.

ex. Solve the inequalities:

$$5\left(\frac{2}{5}x+1\right) < \left(\frac{1}{5}-2x\right)5$$

$$2x+5 < 1-10x$$

$$12x < -4$$

$$x < \frac{-4}{12}$$

$$\boxed{x < -\frac{1}{3}}$$

$$\boxed{\left(-\infty, -\frac{1}{3}\right)}$$

$$2\left[-1 < \frac{3}{2}(2-x) \leq 3\right]$$

Translates into:

$$-1 < \frac{3}{2}(2-x)$$

and

$$\frac{3}{2}(2-x) \leq 3$$

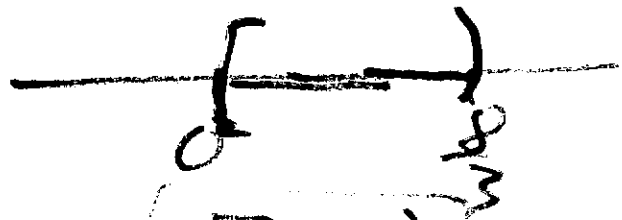
$$-2 < 3(2-x) \leq 6$$

$$-2 < 6-3x \leq 6$$

$$-8 < -3x \leq 0$$

$$\frac{8}{3} > x \geq 0$$

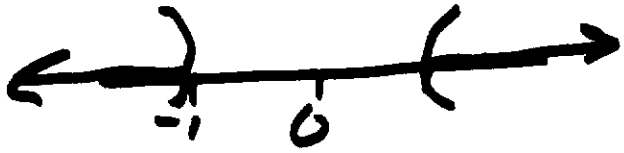
$$0 \leq x < \frac{8}{3}$$



$$\boxed{\left[0, \frac{8}{3}\right)}$$

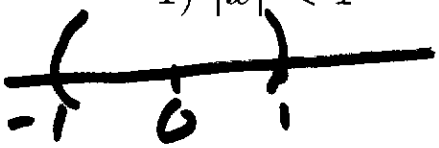
Inequalities Involving Absolute Value

Recall: $|x|$ is the distance from x to 0 on the number line.



<u>ex.</u> Solve:

1) $|x| < 1$



$-1 < x < 1$ also $(-1, 1)$

2) $|x| > 1$

$x < -1$ or $x > 1$

$(-\infty, -1) \cup (1, \infty)$

To Solve an Absolute Value Inequality

shortcut

If a is a nonnegative real number and x is an algebraic expression, then

$|x| < a$ if and only if $-a < x < a$

$|x| > a$ if and only if $x < -a$ or $x > a$

<u>ex.</u> Solve the inequalities:

1) $|x + 2| < -10$ $\emptyset$

pos. neg

2) $|2x + 1| > -5$

all reals

$(-\infty, \infty)$

3) $|x-5| < 2$

(i.e. Find all numbers within 2 units of 5)

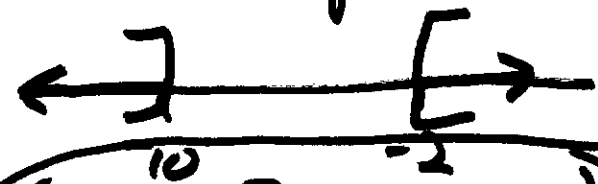
$-2 < x - 5 < 2$

$3 < x < 7$ or $\boxed{(3,7)}$

Isolate the absolute value before starting.

4) $-2\left|3+\frac{x}{2}\right| \leq -4$

$\left|3+\frac{x}{2}\right| \geq 2$



$(-\infty,-10] \cup [-2,\infty)$

$3+\frac{x}{2} \leq -2$ or $3+\frac{x}{2} \geq 2$

$\frac{x}{2} \leq -5$ $\quad$ $\frac{x}{2} \geq -1$

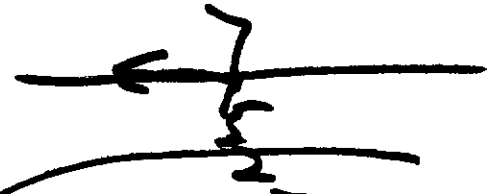
$x \leq -10$ or $x \geq -2$

Applications of Inequalities

<u>ex.</u> Find the domain of x in the expression $\sqrt{5-2x}$.

$5-2x \geq 0$

$-2x \geq -5$



$x \leq \frac{5}{2}$ or $(-\infty, \frac{5}{2}]$

Revenue = how much you sell it for (money you collect)

Cost = your expences

Profit = R − C

ex. 1) Suppose that a company can sell a new product for \$19. If the cost to make the product is \$8 per unit plus a fixed cost of \$88, how many items must the company sell in order to make a profit?

Let x = # of items

R: \$19x

C: \$8x + \$88

P = 19x − (8x + 88) = 11x − 88 = P

we want P > 0

11x − 88 > 0

11x > 88

x > 8

Must sell 9 to make a profit

2) If one week the daily profit from the sale of the items ranged from \$44 to \$110, over what range did daily sales vary that week?

$44 \le 11x - 88 \le 110$

$132 \le 11x \le 198$

$12 \le x \le 18$

[12, 18]

Lecture 6, Part II: Section A.7
Common Algebraic Errors

Errors Involving Parentheses

1. $a - (x - b) \neq a - x - b$

3 terms

2. $(a + b)^2 \neq a^2 + b^2$

3. $\left(\frac{1}{2}a\right)\left(\frac{1}{2}b\right) \neq \frac{1}{2}(ab)$ $\rightarrow \frac{1}{4}ab$

PEMDAS

4. $(3x + 6)^2 \neq 3(x + 2)^2$

Errors Involving Fractions

1. $\dfrac{a}{x + b} \neq \dfrac{a}{x} + \dfrac{a}{b}$

2. $\dfrac{\left(\frac{x}{a}\right)}{b} \neq \dfrac{bx}{a}$

3. $\dfrac{1}{a} + \dfrac{1}{b} \neq \dfrac{1}{a + b}$

4. $\dfrac{1}{3x} \neq \dfrac{1}{3}x$

$\frac{1}{3}x^{-1}$

$\frac{1}{3} \cdot x = \frac{1}{3} \cdot \frac{x}{1} = \frac{x}{3}$

5. $(1/3)x \neq \frac{1}{3x}$

$\frac{1}{x} + \frac{2}{1}$

6. $(1/x) + 2 \neq \frac{1}{x+2}$

Errors Involving Exponents

$y^2 \cdot y^2 \cdot y^2$

$y \cdot y \cdot y y \cdot y y$

1. $(x^2)^3 \neq x^5$

2. $x^2 \cdot x^3 \neq x^6$

$-3^4 = -1 \cdot 3^4$

3. $2x^3 \neq (2x)^3$

4. $\frac{1}{x^2 - x^3} \neq x^{-2} - x^{-3}$

$x+1 \neq (x+y^{-2})y^2$

ex. True or False: $\cancel{\frac{1}{x+y^{-2}} = \frac{y^2}{x+1}}$

$\frac{1}{x + \frac{1}{y^2}}$ = simplify complex fraction

Errors Involving Radicals

1. $\sqrt{5x} \neq 5\sqrt{x}$

2. $\sqrt{x^2 + a^2} \neq x + a$

3. $\sqrt{-x + a} \neq -\sqrt{x - a}$

Errors Involving Dividing Out

1. $\dfrac{a + bx}{a} \neq 1 + bx$

2. $\dfrac{a + ax}{a} \neq a + x$

<u>**ex.**</u> True or False: $\dfrac{2x + y}{2x - y} = \dfrac{x + y}{x - y}$

can cross mult to see that it is false.